

1 ☆ QUESTION: 1 GROUP: Technical SECTION: Programming Ability Mark(s): 2

2 What will be the output of the function given below if called as func(-9)?

```
3 void func(int x)
4 {
5     try
6     {
7         if (x > 0)
8             throw x - 2;
9         else
10            throw x + 5;
11    } catch (int x)
12    {
13        cout << " A " << x;
14    }
15 }
16 int main()
17 {
18     func(-9);
19     cout << " C ";
20     return 0;
21 }
```

A-4-C

1 QUESTION: 2 GROUP: Technical SECTION: Networking Mark(s): 2

2 ☆

What does the "integrity" component of an EISP refer to?

3

☐ Encrypting data during transmission

4

☐ Protecting data from unauthorized access

5

6

☒ Ensuring data is accurate and not altered ✓

7

☐ Making data available to authorized users

8

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1

QUESTION: 3 GROUP: Technical SECTION: Operating Systems Mark(s): 2

2

Which of the below given statement(s) is/are correct about priority scheduling algorithm?

3 ☆

I) The algorithm provides a good mechanism to schedule processes properly as defined.

4

~~II) Static priority keeps changing till the process completes its execution.~~

5

III) Internal priority depends on system limits and memory requirements.

6

☐ Only (III)

7

☒ Only (I) and (III)

8

☐ Only (II) and (III)

9

☐ Only (II)

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1

QUESTION: 4 GROUP: Technical SECTION: RDBMS and SQL Mark(s): 2

2

Given:

3

4 ☆

5

Student_Details			
First_name	Last_name	Age	Subject
Anjali	Bhagvat	12	Maths
Shekhar	Gowda	15	Maths
Rahul	Sharma	10	Science
Rakesh	Roshan	15	History

6

Which of the below given query will display the name of the student with second character 'a' from Student_Details table.

7

☐ SELECT first_name, last_name FROM Student_Details WHERE first_name LIKE 'a%';

8

☒ SELECT first_name, last_name FROM Student_Details WHERE first_name LIKE '_a%';

9

☐ SELECT first_name, last_name FROM Student_Details WHERE first_name LIKE '%_a%';

10

☐ SELECT first_name, last_name FROM Student_Details WHERE first_name LIKE '%_a';

11

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14

1

QUESTION: 5 GROUP: Technical SECTION: RDBMS and SQL Mark(s): 2

2

Which of the queries given below will add a Designation column to an Employee table?

3

☐ Update table Employee add designation varchar(20);

4

☐ Rename table Employee add designation varchar(20);

5 ☆

6

☐ Modify table Employee add designation varchar(20);

6

7

☒ Alter table Employee add designation varchar(20);

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QUESTION: 6 GROUP: Technical SECTION: OOPS & Solid Design Principles Mark(s): 2

2

How does SRP (Single Responsibility Principle) improve the maintainability of a software system?

3

4

☐ By reducing the number of classes.

5

☒ By allowing changes to be made to one part of the class without affecting other responsibilities.

6 ☆

☐ By making each class more complex and capable of multiple tasks.

7

☐ By combining related classes to reduce the amount of code.

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2

QUESTION: 7 GROUP: Technical SECTION: OOPS & Solid Design Principles Mark(s): 2

3

Which of the below given statement(s) is/are true?

4

I) Static members can be applied without creating or referencing any instance of the class.

5

II) A static variable will be shared by all instances of that class which will result in only one copy.

6

☐ Neither (I) nor (II)

7 ☆

☐ Only (II)

8

☒ Both (I) and (II)

9

☐ Only (I)

10

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16

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3

QUESTION: 8 GROUP: Technical SECTION: Networking Mark(s): 2

4

Which layer of OSI model, doeses datagram switching operates on?

5

6

☐ Transport layer

7

☒ Network layer

8 ☆

☐ Data link layer

9

☐ Physical layer

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4

QUESTION: 9 GROUP: Technical SECTION: Programming Ability Mark(s): 2

5

What will come in the blank spaces so that the given program will print the pattern below?

6

7

8

```
  A
 A B A
A B C B A
A B C D C B A
```

9 ☆

10

```
Pattern()
  for i <- 3 until 0
    _____
    write("\n")
    for j <- i until 0
      write(" ")
    end for
    for k <- i till 3
      write(ch++)
    end for
    ch <- ch - 2
    for j <- i till 2
      _____
    end for
```

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4

5

Pattern()

6

for i <- 3 until 0

7

write("\n")

8

for j <- i until 0

write(" ")

end for

9 ☆

for k <- i till 3

write(ch++)

end for

11

ch <- ch - 2

for j <- i till 2

12

end for

13

for k <- i until 0

write(" ")

14

end for

15

end for

16

☒ ch <- 97

17

write(ch)

18

☐ ch <- 65

Previous Question

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5

QUESTION: 10 GROUP: Technical SECTION: Programming Ability Mark(s): 2

6

Function Add(Int N)

7

BEGIN

8

IF (N <= 1)

9

BEGIN

RETURN 0;

10 ☆

END;

ELSE

11

BEGIN

RETURN (N + Add(N - 1) + Add(N - 2));

End;

12

End;

13

What value is returned by the call Add(4) ?

14

☐ 0

15

☐ 4

16

☒ 11

17

18

☐ 9

19

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6

QUESTION: 11 GROUP: Technical SECTION: OOPS & Solid Design Principles Mark(s): 2

7

In the context of the given class diagram for the parking lot system, which of the following statements best illustrates a potential violation of the Interface Segregation Principle (ISP)?

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11 ☆

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☐ If the Car class included methods for managing parking lot functionality, such as adding or removing itself from the lot.

☐ The Ticket class is responsible for managing the issue date and the associated car.

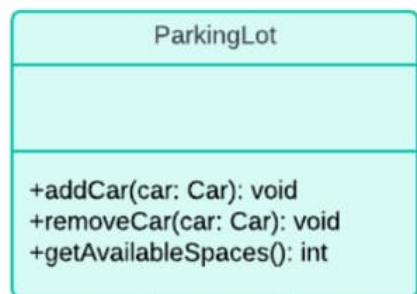
☐ The ParkingLot class provides methods for adding and removing cars, as well as checking available spaces.

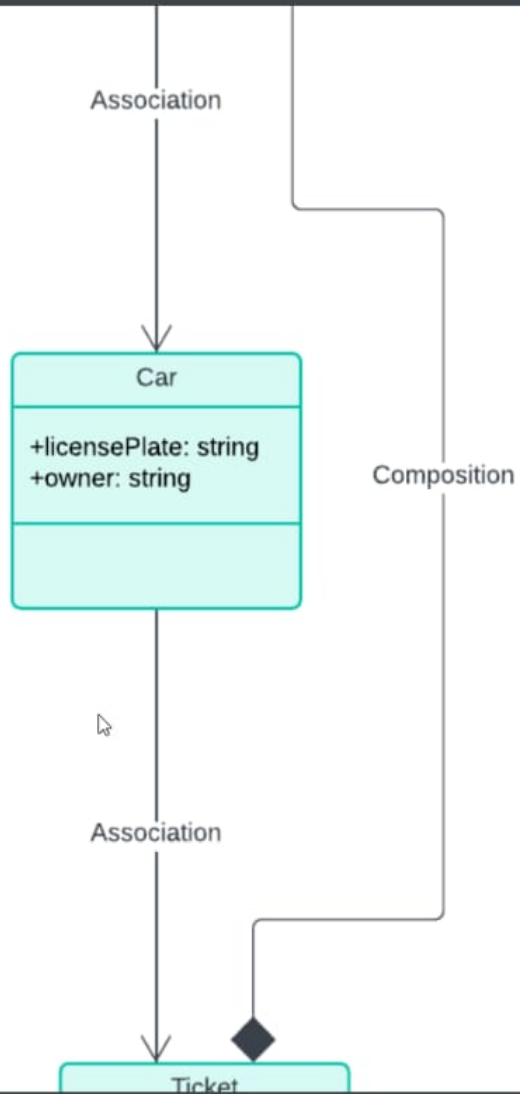
☐ The Car class contains attributes for the license plate and owner information.

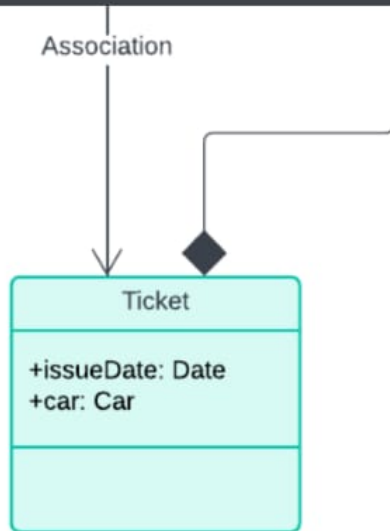
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In the context of the given class diagram for the parking lot system, which of the following statements best illustrates a potential violation of the Interface Segregation Principle (ISP)?







- ☐ If the Car class included methods for managing parking lot functionality, such as adding or removing itself from the lot.
- ☐ The Ticket class is responsible for managing the issue date and the associated car.
- ☐ The ParkingLot class provides methods for adding and removing cars, as well as checking available spaces.
- ☐ The Car class contains attributes for the license plate and owner information.

7

QUESTION: 12 GROUP: Technical SECTION: Data Structures and Algorithms Mark(s): 2

8

Construct a minheap from the nodes given below:

9

44 71 34 68 21 56 42 89 39 75

10

11

Which one of the given options is in the right sub tree of the root?

12 ☆

☒ 42

13

☐ 39

14

☐ 71

15

16

☐ 68

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21

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8

QUESTION: 13 GROUP: Technical SECTION: Data Structures and Algorithms Mark(s): 2

9

10

Children are participating in a hopping competition. To identify the winners the hopping data needs to be organized. To do this deque data structure is used. Consider one deque that has four elements (contains hopping distance as elements) which indicates the old records of hopping competition.

11

2	3	4	5
---	---	---	---

Front

Rear

12

13 ☆

Consider the following hopping status table.

14 ★

SLNo	Hopped Place	Distance
1	X	30
2	X	40
3	Z	
4	Y	38
5	Y	50
6	Z	
7	W	

17

18

Perform the following operations on the deque to organize the new hopping data along with the old records:

19

- 1) Whenever the children hop to place X insert into the rear end.
- 2) Whenever the children hop to place Y insert into the front end.
- 3) Whenever the children hop to place Z delete from the rear end.
- 4) Whenever the children hop to place W delete from the front end.

20

21

22

Identify the sum of elements present in the deque after performing all the given operations to get the total distance covered.

23

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8

Front

Rear

9

Consider the following hopping status table.

Sl.No	Hopped Place	Distance
1	X	30
2	X	40
3	Z	
4	Y	35
5	Y	50
6	Z	
7	W	

10

11

12

13 ☆

14

Perform the following operations on the deque to organize the new hopping data along with the old records:

- 1) Whenever the children hop to place X insert into the rear end.
- 2) Whenever the children hop to place Y insert into the front end.
- 3) Whenever the children hop to place Z delete from the rear end.
- 4) Whenever the children hop to place W delete from the front end.

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Identify the sum of elements present in the deque after performing all the given operations to get the total distance covered.

18

19

☐ 82

20

☐ None of the options given

21

☐ 52

22

☐ 162

23

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end for

end for

☐ ch <- 97
write(ch)

☐ ch <- 65
write(ch-)

☒ ch <- 65
write(ch++)

☐ ch <- 97
write(ch++)

9

QUESTION: 14 GROUP: Technical SECTION: Data Structures and Algorithms Mark(s): 2

10

Construct a min heap from the elements given below:

11

23 55 21 56 46 11 6

12

13

Which one of the element given in options is the in-order predecessor of 11?

14 ☆

☐ 6

15

☒ 23

16

☐ 56

17

18

☐ 21

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10

QUESTION: 15 GROUP: Technical SECTION: Operating Systems Mark(s): 2

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There are four processes operating on the system: P1, P2, P3, and P4. P1 holds resource A and needs B, P2 holds B and needs C, P3 holds C and needs D, and P4 holds D and needs A. What situation describes the current state of the system?

13

☒ All processes are experiencing a deadlock

14

☐ All processes are operating smoothly, there are no problems

15 ☆

16

☐ There isn't a deadlock, although the consumption of resources could be improved

17

☐ There is a deadlock affecting Processes P1 and P2

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11

QUESTION: 16 GROUP: Technical SECTION: OOPS & Solid Design Principles Mark(s): 2

12

If you want to ensure that your system is flexible enough to add new functionality without changing existing code, which SOLID principle should you focus on?

13

14

☒ Open/Closed Principle (OCP)

15

☐ Liskov Substitution Principle (LSP)

16 ☆

☐ Single Responsibility Principle (SRP)

17

☐ Interface Segregation Principle (ISP)

18

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QUESTION: 17 GROUP: Technical SECTION: OOPS & Solid Design Principles Mark(s): 2

In which of the below conditions, can Inheritance be implemented?

- I) If super class has a non-private constructor and subclass is created in the same package.
- II) If super class has public and protected constructors and subclass is created in a different package.
- III) If super class doesn't have any access specifier to a constructor and subclass is created in a different package.

☒ Only (I) and (II)

☐ Only (II) and (III)

☐ Only (I) and (III)

☐ All (I), (II) and (III)

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QUESTION: 18 GROUP: Technical SECTION: OOPS & Solid Design Principles Mark(s): 2

Consider the statements given below:

- I) It wraps the data with the code that manipulates it.
- II) It keeps the data and the code safe from external interference.

To which of the OOPS concepts given in the options do the above statements refer?

☐ Abstraction

☐ Polymorphism

☐ Message passing

☒ Encapsulation

12

QUESTION: 19 GROUP: Technical SECTION: Data Structures and Algorithms Mark(s): 2

13

Consider a stack 'U' of size 5. Sort the below given list in descending order using bubble sort and PUSH the last four elements from 2nd pass to 'U'.

14

24 90 31 22 48 63

15

Which of the given options is the correct order of POPed elements from 'U'?

16

☐ 63 48 22 31

17

☐ 22 63 48 24

18

☒ 31 22 48 63

20

☐ 24 48 63 22

21

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QUESTION: 20 GROUP: Technical SECTION: OOPS & Solid Design Principles Mark(s): 2

According to the Dependency Inversion Principle (DIP), high-level modules should depend on _____ rather than concrete implementations.

☐ Direct instances

☒ Abstractions

☐ Static classes

☐ Constructors

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QUESTION: 21 GROUP: Technical SECTION: RDBMS and SQL Mark(s): 2

Consider the Emp_details table given below:

Emp_Details		
Emp_id	Name	Sal
10	Rinky	9000
20	Rahul	11000
30	Sarita	7000
40	Akash	-
50	Rakesh	20000
60	Rakesh	15000

Which of the query/queries given below will retrieve third, fourth records from Emp_details table?☐ SELECT * FROM (SELECT ROWNUM, Emp_details.* FROM Emp_details) WHERE ROWNUM IN(3,4);☒ SELECT * from (SELECT ROWNUM R, Emp_details.* from Emp_details) where R IN(3,4);☐ SELECT * FROM Emp_details WHERE ROWNUM IN(3,4);☐ SELECT ROWNUM R, Emp_details.* FROM Emp_details WHERE R IN(3,4);

12

Consider the table given below:

13

14 ★

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21

22 ☆

23

24

25

Students		
Id	Stud_name	age
10	Rajesh	25
20	Kapil	28
30	Jony	20
40	Jony	30
50	Mohit	40

What will be the output of the query given below?

```
SELECT ID, AVG (age)
```

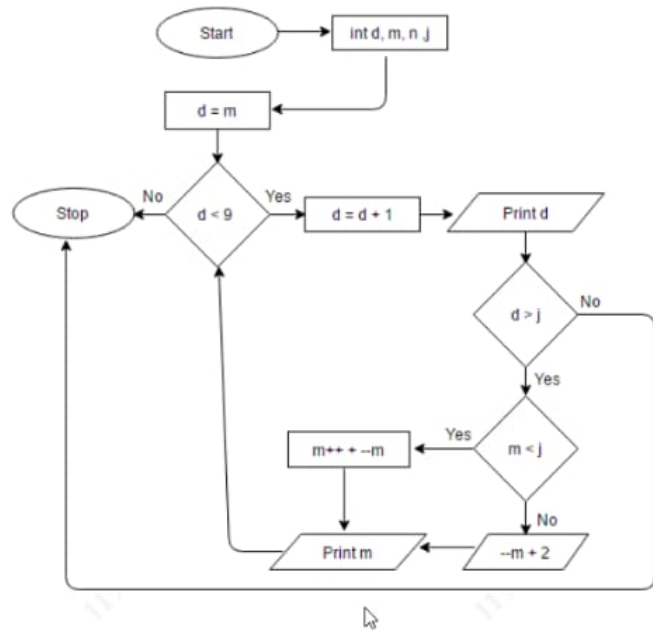
```
FROM students
```

```
HAVING AVG(age) > 30
```

```
GROUP BY ID;
```

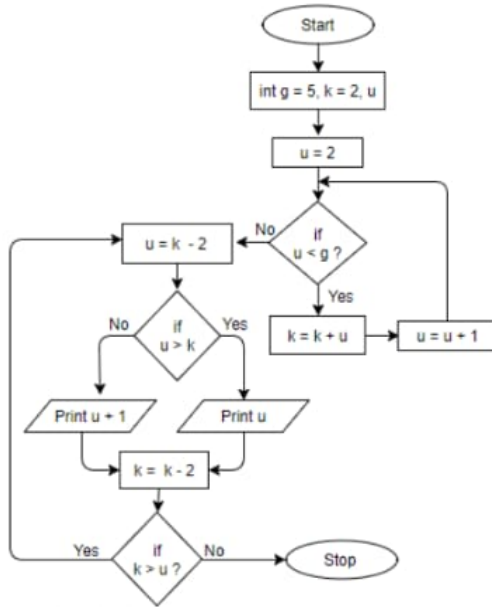
☐ 40,50☐ 30,20☐ 30,40☒ 50,40

10 Which one of the statement given in options is correct about the flowchart given below where $d = 4$, $n = 1$, $m = n + d$, $j = d + 2$?



- 21 ☐ It will print 57.
- 22 ☐ It will enter into infinite loop.
- 23 ☆ ☒ It will print the value of 'd' and comes out of the program.
- 24 ☐ It will print the value of "-m + 2".

What will be the output of the flowchart given below?



☒ 10

☐ 11

☐ 12

☐ 9

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26 ☆

QUESTION: 26 GROUP: Coding SECTION: Coding 1 Mark(s): 20

27

Shilpa wants to go to a movie with her friends. There are totally N friends including Shilpa. M different offers were available based on the number of tickets needed. A different number of tickets are sold at different group booking rates. For example, a single ticket may cost 10 rupees, two tickets may cost 15 rupees together, and 3 maybe 40 rupees together.

The ticket seller always introduces M offers which are always equal to the number of friends N . Ticket seller would like to maximize the revenue from Shilpa. Write a program to determine how can he do it given M and the M costs for k tickets together, $k = 1, 2, \dots, N$.

Read the input from STDIN and print the output to STDOUT. Do not write arbitrary strings while reading the input or while printing, as these contribute to the standard output and testcases will fail.

Constraints: $1 < N < 1000$ $M = N$ **Input Format:**

The first line of input consists of N , the number of friends, and the number of ticket possible combinations available M .

The second line has N integers, which are costs for k tickets together, $k = 1, 2, \dots, N$.

Output Format:

The output consists of a single integer S , the maximum revenue the ticket seller can earn.

Sample Input1:

```
3
10 15 20
```

Sample Output1:

```
30
```

Explanation:

If picks the first offer of 10 rupees per ticket then the total ticket cost would be $10+10+10 = 30$

If picks the second offer of 2 tickets costs 15 other one costs 10 rupees then the total cost would be $10+15 = 25$

If picks the third offer of 3 tickets costs 20 rupees then the total cost would be $= 20$

If the ticket seller suggests the first offer then he would be more beneficial. So that is printed as output

Sample Input2:

```
3
20 70 110
```

Sample Output2:

```
110
```

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26 ☆

Explanation:

If picks the first offer of 20 rupees per ticket is chosen then the total ticket cost would be $20+20+20 = 60$

If picks the second offer of 2 tickets costs 70 and the other ticket costs 20 rupees then the total cost would be $= 20+70 = 90$

If picks the third offer of 3 tickets costs 110 rupees then the total cost would be $= 110$

If the ticket seller suggests the third offer then he would be more beneficial. So that is printed as output

27

Current Code

Previous Versions

C++

```
1 #include <iostream>
2 using namespace std;
3
4 int maximumRevenue(int n,int* arr)    // n is the number of friends and arr is the array of n integers,which are costs for k tickets,k=1,2,----n
5 {
6     int max = 0;
7
8     // WRITE YOUR CODE HERE
9
10    return max;
11
12 }
13
14 int main() {
15     int n;
16     cin>>n;
17     int arr[n];
18     for (int i = 0; i < n; i++)
19         cin>>arr[i];
20     cout<<maximumRevenue(n,arr);
21     return 0;
22 }
23
```

Previous Question

Next Question


```
int maxRevenue(int N, int *arr) {  
    vector<int> dp(N + 1, 0);  
  
    for (int i = 1; i <= N; ++i) {  
        dp[i] = 0;  
        for (int k = 1; k <= i; ++k) {  
            dp[i] = max(dp[i], arr[k - 1] + dp[i - k]);  
        }  
    }  
  
    return dp[N];  
}
```

ROD CUTTING

26 QUESTION: 27 GROUP: Coding SECTION: Coding 2 Mark(s): 30

27 ☆

After delivering a lecture on rainwater harvesting on Environment day, the Director of the Institute asks students to construct a recharge pit that allows rainwater harvesting on the ground in the backyard. The recharge pit is made up of $N \times M$ cells, where N is the number of departments and M is the number of students from each department.

Each student has to raise or lower the soil level in the cell allotted to him/her to harvest rainwater. The students work as a team and decide the level of soil in each cell considering the slope of the land, porosity, and texture of the soil. After completing the work, the final layout is represented as integers where 0 represents ground level, -ve represents dug up areas and +ve represents raised areas. The integer value represents the height or depth of the soil level.

Water can be trapped in a cell only if all the four sides of a cell are above the cell level. You are given the elevation map which shows the level of soil in each cell.

Write a program to find the quantity of water that can be trapped in the recharge pit.

Read the input from STDIN and print the output to STDOUT. Do not write arbitrary strings anywhere in the program, as these contribute to the standard output and test cases will fail.

Constraints:I) $3 \leq N \leq 100$ II) $3 \leq M \leq 100$ **Input Format:**

The first line of input contains two integers N and M , separated by a single white space.

The next N lines of input contain M integers denoting the elevation-map of the recharge pit, with integers separated by single white spaces.

Output Format:

The output should contain the amount of rainwater that can be trapped in the pit.

Sample Input 1:

```
3 3
2 2 2
2 -1 2
2 2 2
```

Sample Output 1:

```
3
```

Explanation 1:

$N = 3, M = 3$.

The pit elevation map is given as follows:

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26

Sample Output 1:

3

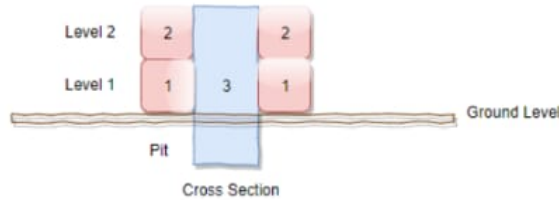
27 ☆

Explanation 1: $N = 3, M = 3.$

The pit elevation map is given as follows:



Top View



It can be seen that the cells with values 2 act as the boundary and only in the cell with value -1, water will get trapped. As the difference between the two is 3 and the cells have unit area, the quantity of water that can be stored is 3, which is printed as the output.

Sample Input 2:

4 5

2 1 1 2 1

3 -2 -2 1 -2

2 -3 -4 1 0

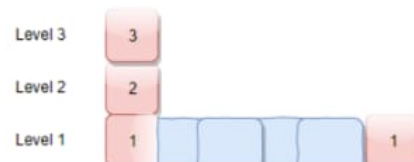
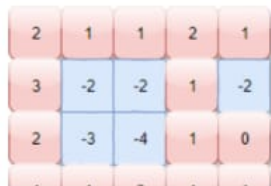
1 1 2 1 1

Sample Output 2:

17

Explanation 2: $N = 4, M = 5.$

Elevation map is represented by

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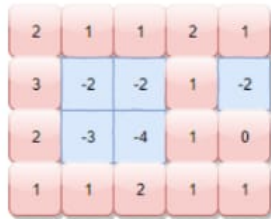
26

Explanation 2:

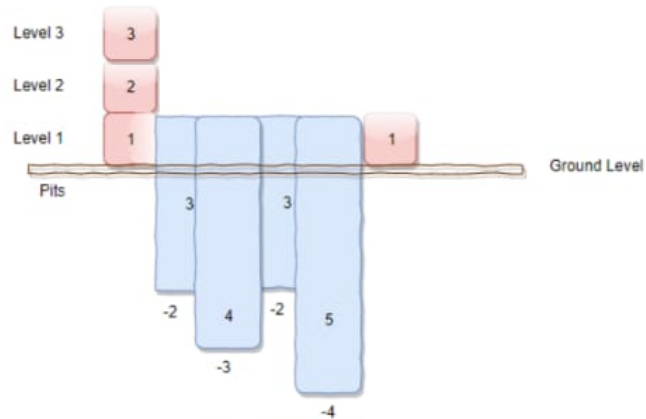
N = 4, M = 5.

Elevation map is represented by

27 ☆



Top View



Cross Section View

Pits are represented by -ve numbers, +ve number represents raised areas and 0 represents the ground level.

The shaded cells act as the boundary for water trapping and water gets trapped in area inside the boundary. Since the boundaries have different height levels, water can be trapped only till the level of 1.

Water trapped in pit represented as $-2 = 1 - (-2) = 3$.

Water trapped in pit represented as $-3 = 1 - (-3) = 4$.

Water trapped in pit represented as $-4 = 1 - (-4) = 5$.

water trapped in pit represented as -2 at the border can also hold 2 units of water hence =2

There are 2 pits marked as -2.

Total amount of water = $2*3+4+5+2=6+4+5+2=17$.

Current Code

Previous Versions

C++

```
1 #include <iostream>
2 using namespace std;
3
4 int trapRainWater(int** heightMap, int n, int m) // heightmap is an array of integeres with n rows and m cols.
5 {
```

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End Test

26

CROSS SECTION VIEW

27 ☆

Pits are represented by -ve numbers. +ve number represents raised areas and 0 represents the ground level.

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Water trapped in pit represented as -3 = $1 - (-3) = 4$.

Water trapped in pit represented as -4 = $1 - (-4) = 5$.

water trapped in pit represented as -2 at the border can also hold 2 units of water hence =2

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Total amount of water = $2*3+4+5+2=6+4+5+2=17$.

Current Code

Previous Versions

C++

```
1 #include <iostream>
2 using namespace std;
3
4 int trapRainWater(int** heightMap, int n, int m)    // heightmap is an array of integeres with n rows and m cols.
5 {
6     int totalwater = 0;
7
8     // WRITE YOUR CODE HERE
9
10    return totalwater;
11 }
12
13 int main()
14 {
15     int N, M;
16     int** mat;
17     cin >> N >> M;
18     mat = (int**)malloc(sizeof(int*)*N);
19     for(int i=0; i<N; i++)
20         mat[i] = (int*)malloc(sizeof(int)*M);
21     for(int i=0; i<N; i++)
22         for(int j=0; j<M; j++)
23             cin >> mat[i][j];
24     cout << trapRainWater(mat, N, M);
25     return 0;
26 }
27
```

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Next Question

End Test

```

4 void dfs(int r, int c, int n, int m, int **mat, vector<vector<bool>> &vis, int &ans){
5
6     vis[r][c] = true;
7     ans += -mat[r][c];
8     int deli[4] = {0, 1, 0, -1};
9     int delj[4] = {1, 0, -1, 0};
10
11     for(int i=0 ; i<4 ; i++){
12         int nx = deli[i] + r;
13         int ny = delj[i] + c;
14
15         if(nx>=0 && nx<n && ny>=0 && ny<m && !vis[nx][ny]){
16             if(mat[nx][ny] < 0){
17                 dfs(nx, ny, n, m, mat, vis, ans);
18             }
19         }
20     }
21 }
22 }
23
24 int trapRainWater(int **mat, int n, int m){
25
26     vector<vector<bool>> vis(n, vector<bool>(m, false));
27     priority_queue<pair<int,pair<int,int>>, vector<pair<int,pair<int,int>>>, greater<pair<int,pair<int,int>>> > pq;
28
29     int ans = 0;
30
31     for(int i=0 ; i<n ; i++){
32         for(int j=0 ; j<m ; j++){
33             if(i==0 || i==n-1 || j == 0 || j == m-1){
34                 if(mat[i][j] < 0){
35                     dfs(i, j, n, m, mat, vis, ans);
36                 }
37             }
38         }
39     }
40

```



```

41     for(int i=0 ; i<n ; i++){
42         for(int j=0 ; j<m ; j++){
43             if(i==0 || i==n-1 || j == 0 || j == m-1){
44                 pq.push({mat[i][j], {i, j}});
45                 vis[i][j] = true;
46             }
47         }
48     }
49
50
51     int deli[4] = {0, 1, 0, -1};
52     int delj[4] = {1, 0, -1, 0};
53
54     while(!pq.empty()){
55         auto it = pq.top();
56         pq.pop();
57         int ht = it.first;
58         int x = it.second.first;
59         int y = it.second.second;
60
61         for(int i=0 ; i<4 ; i++){
62             int nx = x + deli[i];
63             int ny = y + delj[i];
64
65             if(nx>=0 && nx<n && ny>=0 && ny<m && !vis[nx][ny]){
66                 ans += max(0, ht - mat[nx][ny]);
67                 pq.push({max(ht, mat[nx][ny]), {nx, ny}});
68                 vis[nx][ny] = true;
69             }
70         }
71     }
72
73
74     return ans;
75 }

```